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High Performance Processors and Systems

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Name
Last Name

Problem 1 (30%)	
Question 1 (20%)	
Question 2 (30%)	
Question 3 (20%)	
Total (100%)	

Problem 1

Assume that the following code is executed on a CPU with TOMASULO and with the following units:

- 1 LOAD/STORE unit with Latency equal to 1
- 2 MULT units with latency equal to 15
- 3 ADD/SUBD unit with latency equal to 2

And the following information: First load 10 clock cycles (cache miss),
Following loads 1 clock cycle (hit), R1 = 30

Show the computation steps of the following lines of code:

```

Loop:    LD F0 0 R1
         ADD F6 F4 F0
         MULTD F10 F6 F0
         ADD F6 F0 F10
         ADD F4 F10 F6
         SUBI R1 R1 #15
         BNEZ R1 Loop

```

[illegible]

Tables

[illegible][illegible][illegible]

[illegible][illegible][illegible]

[illegible][illegible][illegible]

[illegible][illegible][illegible]

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Question 1

1. Explain the difference between implicit and explicit register renaming.
2. Can an architecture with Tomasulo scheduling use explicit register renaming? Explain.
3. How precise exceptions can be guaranteed with implicit or explicit register renaming?

Question 2

Consider a 2 processors systems with shared memory that adopts the MSI **write-invalidate write-back protocol**.

1. Define which are the states in which each cache block can be and explain how this protocol ensures coherence;
2. Explain what happens under this protocol when the following operations are performed by the two processors P1 and P2. Assume a shared variable stored in address A1 of the memory with initial value 1. Consider initially that the cache blocks are **invalid**.
 - a. P1 reads A1 (this line has been provided)
 - b. P2 writes 2 to A1
 - c. P1 reads A1
 - d. P1 writes 4 to A1
 - e. P2 writes $3*[A1]$ to A1

Operation	Type of cache action	Cache content of P1	Status of the cache block at the end of the operation	Cache content of P2	Status of the cache block at the end of the operation	Memory content for location A1
P1 reads A1	Read miss	1	S	?	I	1
P2 writes 2 to A1						
P1 reads A1						
P1 writes 4 to A1						
P2 writes $3*[A1]$ to A1						

Question 3

Explain how the following elements represent a limitation to ILP.

- a. Instructions window size
- b. Number of registers
- c. Branch prediction
- d. Memory address alias analysis

